

“As the world talks about green products, power supply design has gained a lot of wattage”



While talking about analogue design and business, evolution of electronics power supply and latest developments in power supply ICs, Balu Balakrishnan, president and CEO, Power Integration, and winner of 'Discover Award for Technological Innovations' and 'TechAmerica Innovator Award,' tells Shweta Dhadiwal Baid of EFY Bureau that energy efficiency is the key growth driver

■ **BALU BALAKRISHNAN**
PRESIDENT AND CEO, POWER INTEGRATION

Everyone is talking about 'digital' in the technology domain. What are your views on analogue design and business? In all this, where does the power supply design stand?

Analogue business is here to grow, despite all the advancements in digital. There are very few companies like Maxim, Linear Technology and National Semiconductor that are purely into analogue business and they are enjoying much higher profit margins than digital.

Every interface or interaction of an electronic device with the rest of the world is in the analogue domain, so understanding analogue concepts is very important. For example, a mobile phone is made with the most high-end digital technology and digital ICs, but without analogue nothing happens. The sound that comes from your speaker is analogue. The battery uses an analogue circuit in the charger for charging. In my opinion, analogue engineers are into a very stable business, unlike the volatile digital market where technology changes very rapidly.

As the entire world talks about designing green or energy-efficient products, power supply design has gained a lot of wattage with a number of innovations happening in the domain. When you are dealing with various ranges of voltages and currents, the efficiency brought in by a power supply plays an important role.

What are the key developments happening in electronic power supplies and semiconductors?

In order to drive any electrical or electronic gadget, you need a power supply which is actually a power adaptor. In 1970s, all the power supplies were built using linear trans-

formers. Later, when high-voltage transistors came into the market, they could actually build discrete electronic power supplies which were smaller and lighter but very complex. In 1994, the complexity was dramatically reduced by integrating these components in the form of a semiconductor IC.

Energy efficiency is the major trend in power supply design. The transformers are phasing out because these typically show only 30 per cent efficiency. So in the next couple of years you will not be able to buy transformers easily. Technologies like 'EcoSmart' help create more and more energy-efficient devices.

Another important development is the standby power requirement for external power supply, which has become mandatory in all of Europe, Australia, New Zealand and China. Associations are working to extend these standards to cover newer products.

Creating smaller, lighter and efficient power supplies is what everyone is talking about. This has led to a lot of integration in semiconductors. Many features and functionalities are getting integrated onto a single chip, except for the functions related to safety like electromagnetic interference (EMI) filters and energy storage transformers. At the same time, the size of the chip and price are important. Integration also reduces the power supply design time because it is a lot simpler to design with minimal components. Most of the complexity is inside the chip.

How does EcoSmart technology help in bringing more efficiency to the power supply?

A power supply is normally designed to work with maximum efficiency at a particular load. That is why you